Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

ASEA

Project / Solution Name

Path.AI: find your path in the AI era

Name (Leader)

Almat Sakiidulla

Email & Phone (Leader)

sat200707@mail.ru | +77072203626 (WhatsApp)

University Affiliation(s)

Xiamen University Malaysia

Live Prototype URL

https://path-ai-asea.replit.app

Code Repository URL (public / access granted to judges)

https://github.com/Almatikooo/Competition_Repo

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

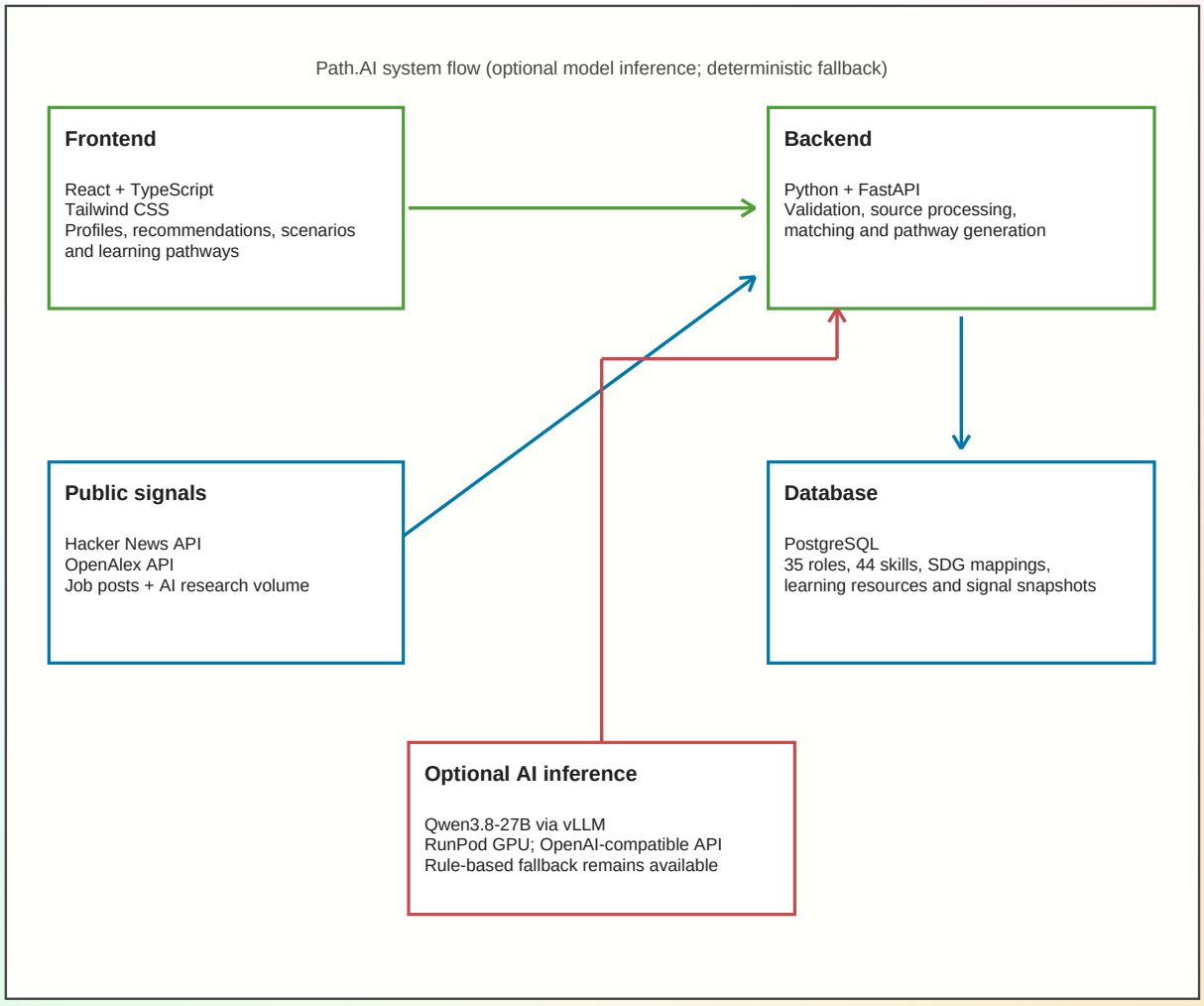
Path.AI uses a modular architecture separating user interface, backend processing, data collection, and AI inference.
1. Frontend: React, TypeScript, and Tailwind CSS single-page interface for profiles, career recommendations, scenario comparisons, and learning pathways.
2. Backend: Python and FastAPI providing REST endpoints for validation, source processing, matching, and pathway generation.
3. Database: PostgreSQL structured storage for 35 catalogued roles (O*NET-SOC), 44 skills, SDG mappings, learning resources, and dated signal snapshots.
4. Data ingestion: public signals retrieved from Hacker News API (job posts) and OpenAlex API (AI research volume covering 12 capabilities).
5. AI inference: open-weight Qwen3.8-27B model hosted via vLLM on RunPod GPU infrastructure, accessed via an OpenAI-compatible API for skill extraction and readable explanations. Deterministic fallback logic ensures full functionality if model inference is disabled.
Block diagram: Appendix A (page 2a).

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Appendix A. System block diagram

Referenced in Section 2.1. Arrows show data flow. The model service is optional; rule-based fallback remains available. PostgreSQL stores catalog data and dated signal snapshots.



2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12 / TypeScript	Core backend language / frontend application logic	PSF License / MIT
React / Tailwind CSS	Responsive user interface component layer and styling	MIT
FastAPI / Uvicorn	Backend REST API framework and ASGI web server	MIT / BSD-3-Clause
Pydantic / httpx	Input validation, schemas, and async HTTP requests to external APIs and model endpoint	MIT / BSD-3-Clause
python-multipart / pypdf / pytest	CV upload handling, PDF text extraction, and automated test suite	Apache-2.0 / BSD-3-Clause / MIT
vLLM / Roboto font	Qwen3.8-27B model server on RunPod / primary web typography (Google Fonts)	Apache-2.0 / SIL OFL 1.1

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Hacker News Search API (Algolia)	Fetching "Who is hiring?" threads and comments for aggregate skill-mention signals	Free, public API
OpenAlex Works API (OurResearch)	Retrieving publication metadata and volume across 12 AI research capabilities	Free, public API (CC0)
vLLM Inference Endpoint (RunPod)	OpenAI-compatible endpoint serving Qwen3.8-27B for CV skill extraction and explanations	Sponsor-provided (RunPod credit)
Google Fonts API	Delivering web typography assets (Roboto font)	Free

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Deployments	Primary application hosting, development, execution, and publication of live prototype	Replit Core (sponsor-provided); region not specified
RunPod GPU Pod	GPU hosting for vLLM inference engine running Qwen3.8-27B-FP8	Secure Cloud (sponsor credit, 1x GPU); region not specified
Hugging Face Hub	Repository host for downloading open pre-trained model weights (Qwen3.8-27B-FP8)	Free, public model hub
GitHub	Version control, source code management, documentation, and public code repository	Free public repository

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Qwen3.8-27B-FP8	Alibaba Qwen Team / Hugging Face	Pre-trained model served via vLLM on RunPod GPU for CV parsing and text explanations	Apache-2.0
Path.AI Scenario & Matching Logic	Self-built (Team ASEA)	Deterministic algorithms for numerical scores, skill fit, scenario projections, and catalogue ranking	Open-source (repository)

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
HN "Who is hiring?" Threads	hn.algolia.com (11,583 postings across 36 threads)	Public (aggregate counts only)	Public API terms	N
OpenAlex AI Research Activity	openalex.org (12 AI capabilities metadata)	Public	CC0 1.0	N
O*NET-SOC Occupation Codes	onetonline.org (35 catalogued roles)	Public	CC BY 4.0	N
WEF Future of Jobs (2023/2025)	World Economic Forum / ILO	Public research baselines	Cited with attribution	N
Open Learning Resources	Curated course catalog (Carpentries, open educational content)	Public learning materials	CC BY 4.0, MIT, OGL v3.0, CC0	N
Synthetic Demo Personas	Self-created (5 fictional profiles: Aisyah, Wei Jie, Kavitha, Hafiz, Nurul)	Synthetic / fictional	Open-source	N (no real personal data)

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Claude Code (Anthropic)	Development environment assistance, boilerplate code generation, and initial document drafting	Moderate
ChatGPT / Copilot	Code auto-completion, refactoring, and debugging FastAPI endpoints	Minor

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

All AI-assisted outputs, code snippets, and documentation drafts were thoroughly reviewed, tested, and validated by human team members prior to integration. Code functionality is verified through automated unit testing (pytest). Numerical forecasts, matching algorithms, and catalogue selections are handled deterministically to prevent hallucination. The team accepts full responsibility for the accuracy, validity, and integrity of all project materials.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Replit Core Platform	Application development, testing, execution, and publication of the live prototype
RunPod	GPU Compute Credits ($25)	Hosting model inference via vLLM for Qwen3.8-27B to perform skill extraction and explanation formatting

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit is essential for hosting and publishing the live prototype as required by the hackathon.

RunPod is non-essential and easily replaceable; the core solution operates deterministically. If GPU inference is unavailable, Path.AI seamlessly falls back to pre-defined rule-based skill detection and structured templates without affecting core career matching or pathway functionality.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Path.AI is designed with strict privacy guardrails and relies exclusively on synthetic demo profiles and open labor market signals. The application does not collect or store personal user data or real CVs. Uploaded CVs for demonstration are processed strictly in-memory for skill extraction and are discarded immediately. User interaction preferences stay entirely within the user's local browser storage. API keys and secrets are securely configured via environment variables and excluded from the version-controlled repository.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

1. Model hallucination: Language models may produce inaccurate details. Mitigation: LLM output is strictly bounded by deterministic schemas and guardrails; core scoring and catalog selections remain 100% deterministic.

2. Signal and data bias: Hacker News postings lean heavily toward tech and global roles. Mitigation: Non-tech roles are clearly labeled with confidence bounds, and fairness tests ensure demographic neutrality in ranking.

3. API and infrastructure availability: Third-party or GPU outages. Mitigation: Local dated signal caching and fallback rule-based logic guarantee full system operation during network disruptions.

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.

☑ Any use of official SDC sponsor resources has been declared in Section 5.

☑ The prototype described here is accessible to judges through the links provided in Section 1.

☑ All fields in this template have been completed within the word limits stated.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Almat Sakiidulla

24 October 2026